

Mine Sampling

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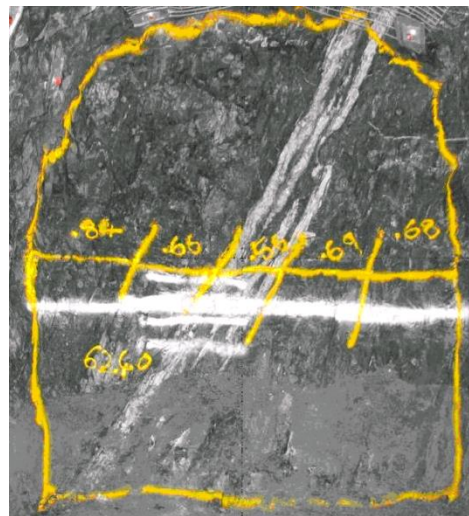
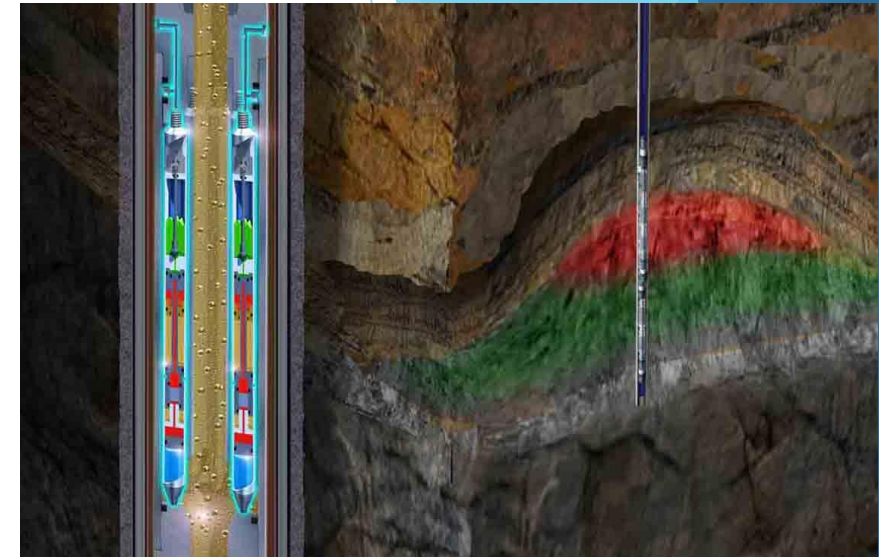
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Contents

- ▶ Definition,
- ▶ Purpose and scope,
- ▶ Methods: groove/channel sampling; Chip sampling; grab sampling; Bulk sampling;
- ▶ Application of statistical methods in sampling

- ▶ Sample is the material collected from mineral deposit to represent the formation with regard to quantities and quality of useful mineral.
- ▶ Sampling defined as: "The operation of removing a part convenient in size for testing, from a whole which is of much greater bulk, in such a way that the proportion and distribution of the quality to be tested (e.g. specific gravity, metal content, recoverability) are the same in both the whole and the part removed (SAMPLE)."
- ▶ Sampling procedures cover the practice of selecting representative quantities of test material in the field, to evaluate bulk materials.



- ▶ Assay is a chemical analysis that determine the proportion of metals in ore
- ▶ **The objectives of the sampling is to**
 - ▶ Determine the average grade of orebody and to estimate the quantities of the reserve.
 - ▶ Quantity of ore present in order to estimate future profit to be derived from the sale of recoverable minerals.
 - ▶ To correlate the different parts of the orebody
 - ▶ In day to-day marking to maintained the proper grade or ROM grade and also unpayable mining
 - ▶ For proper blending to meet the required mill/concentrator grades.

Thus with respect to the samples, it is very important that the sample be representative. In reality, the condition that the sample be completely representative of the whole as regards all aspects except bulk, is never fulfilled when heterogeneous materials are sampled. For instance, you can never get a perfectly representative sample of water from Lake Superior. You may even ask what good it is or what purpose such a sample would serve. It is important, before collecting a sample, to decide what the purpose of the sample would be, and then plan the sampling procedure accordingly so that it will be sufficiently representative for your purposes.

▶ Various stages of mining

- ▶ Prospecting
- ▶ Explorations
- ▶ Planning and design
- ▶ Development
- ▶ Production
- ▶ Reclamation and rehabilitations

▶ Factors to be considered in designing a sampling campaign for mineral property evaluation

- ▶ Objectives of sampling-prospecting,---
- ▶ Population to be sampled
- ▶ Data to be collected
- ▶ Choice of sampling unit to adequately represent the population
- ▶ Organization of field data
- ▶ Cost involved
- ▶ Type of minerals
- ▶ Amount of money available
- ▶ Accessibility of various parts of the mineral deposit
- ▶ Avoid over precision to reduce cost

Classification of sampling methods

Place of collection

- ▶ Dump sampling
- ▶ Bore hole sampling
- ▶ Stope sampling
- ▶ Development sampling

Mode of collection

- ▶ Groove/channel sampling
- ▶ Chip sampling
- ▶ Grab sampling
- ▶ Bulk sampling

On the basis of purpose

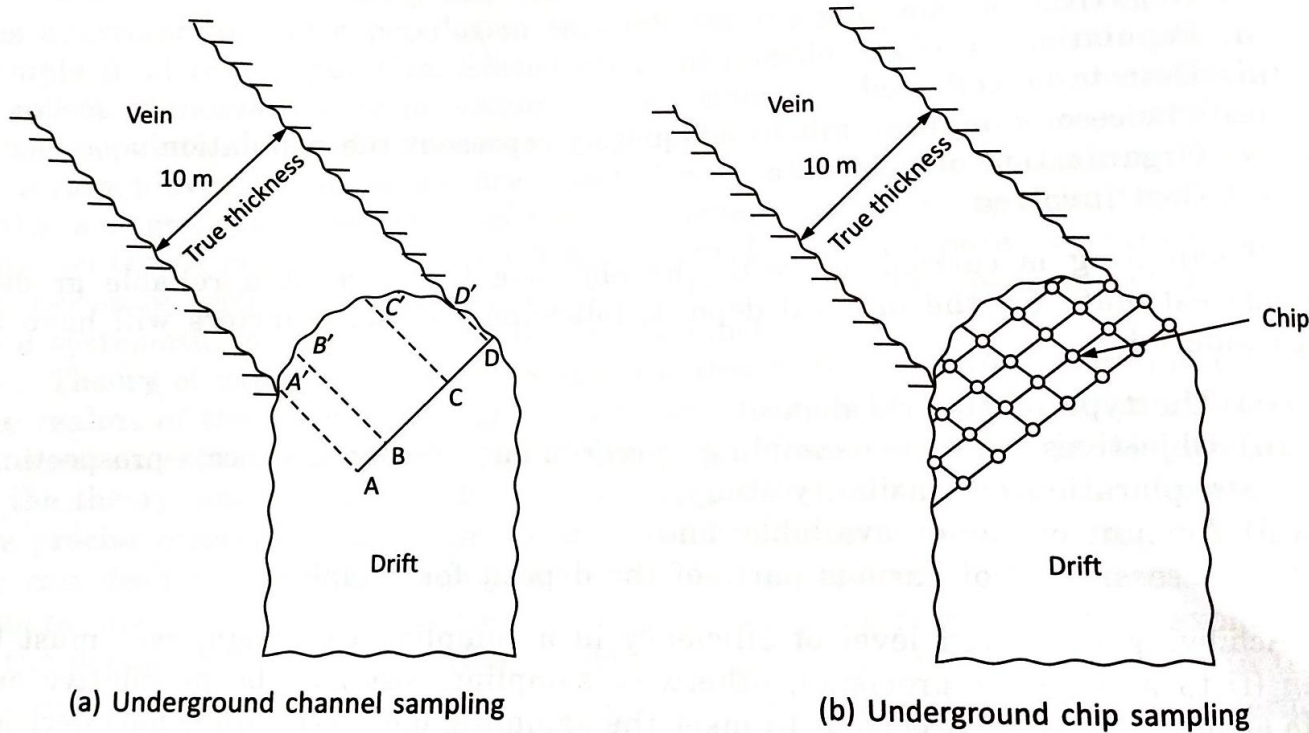
- ▶ Domestic sampling
- ▶ Independent sampling

In a laboratory before performing a size analysis, chemical analysis, or any other determination of material properties, it is necessary to take a representative sample from the lot being analysed.



Channel and Chip Sampling

In channel sampling, a groove of uniform width and depth (5–10 cm) is cut across the exposed ore, preferably in the direction normal to the direction of trend or stratification. Before the sampling groove is cut, the surface to be sampled is cleaned to remove dust and soluble particles, etc. Also the surface is smoothened and, whenever practicable and desirable, a thin layer of the exposed ore is removed to avoid inclusion of weathered ore in the sample.



Channels should generally be around 5 cm to 10 cm wide and 2.5 cm **deep** to yield a **sample** of between 3–6 kg/m



(A)



(B)



(C)



(D)



Efficiency of the campaign depends heavily on the ability to cut neat, uniform and well-defined channels across the mineral exposure. All the cuttings and dust from the channel are then carefully collected. Normally, the channel is divided into sections of 1 m to 2 m for a massive deposit [Figure 4.1(a)]. For a more homogeneous massive deposit, the sections may be greater in length, while for heterogeneous mineralization, the channel sections may be of 30 cu to 60 cu in length. Keeping the channel samples bias free is a formidable task. If visual differentiation is possible, it is preferable to sample layers of different hardness or value separately.

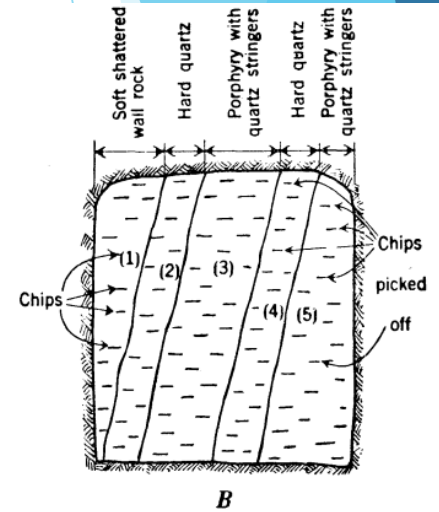
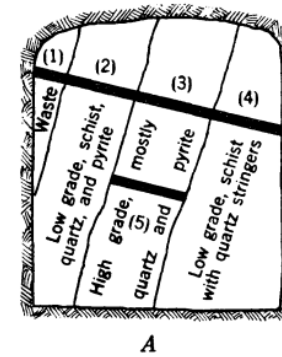


FIGURE 7.—Examples of face sampling. *A*, Channel sampling; numbers indicate separate samples 3 and 5 averaged for high-grade band. *B*, Pick ("chip") sampling; numbers indicate separate samples.

In hard ore-bodies, it is often difficult to cut grooves of uniform dimensions. Under such circumstances, if the mineralization is uniform, **chip sampling** is often used as a substitute for channel samples. Chip sampling is carried out by breaking off small, equal-sized pieces uniformly distributed across the mineral zone. Sampling bias observed in chip sampling, however, is more than the bias normally observed in channel sampling under similar situation. In order to reduce bias, samples of equal size are taken from uniformly distributed grade points [see Figure 4.1(b)].

Grab Sampling or Muck Sampling

Geologists in some underground mines collect grab-samples from broken ore as a method of grade control. It is often known as muck or broken rock sampling. Generally, the goal of grab sampling is to try and reconcile the mined grade at the ore source to the predicted head grade (Holmes, 2010). Additionally, the samples may be used to verify prior *in situ* estimates of grade such as those derived from face samples, sludge holes or diamond drilling. It is often used because of access issues (e.g., non-entry stopes), safety (e.g., to avoid unsupported backs) or lack of other sample data. On surface, grab sampling is often used to monitor the grade of stock piles prior to blending and/or feeding to the plant.



Grab sampling involves collecting a large sample (or series of smaller samples that are later combined) from the muckpile at a face or drawpoint, or from tram cars or trucks transporting the ore from these points.

Samples are generally collected either by hand or shovel by a geologist, field technician or miner.

Non-scientific protocols usually define a sample mass and number of samples per blast or per truck, etc. Time pressures and carelessness can lead to missed samples, for example where only one out of six tram cars is sampled, or where no trams were sampled.

The mass of individual samples is usually in the range of 1 kg to 5 kg. The sample mass is limited in practical terms by health and safety issues, as well as by the capacity of the laboratory to process the samples within a given time.

Realistically, sample sizes required to be representative of broken rock piles are likely to be in the range of 100 kg to 20 t or more.



Muck sampling is normally carried out when composition of run-of-mine (ROM) needs to be ascertained with greater precision. The procedure usually follows a predetermined plan to take a certain bulk of mineral at intervals—either time or distance—to be used for estimating the grade or volume of ore. Samples may be made by collecting a portion from every blast continuously, or from shovels or mine-cars. For prospecting work, samples may also be collected from a series of pits or a number of trenches.

Bulk Sampling

A bulk sample is a large rock sample, frequently hundreds of tons, that is selected to be representative of an ore-body. Bulk samples are used mostly to determine metallurgical characteristics.

Bulk sampling is an essential part of exploration and evaluation of ores in massive deposits like iron ore. It is undertaken when a pilot plant test is to be done to determine metallurgical characteristics. Large samples are required to check the accuracy of the drill samples and to obtain the information on the crushing and grinding characteristics of the ore as well as for use in detailed and large-scale metallurgical tests.

While bulk sampling may be of importance in iron ore, for other ore bodies, high quality drill samples may be sufficient to determine the beneficiation characteristics. In areas where the accumulated knowledge regarding the expected natural ore characteristics is not well known, bulk sampling may be required for all ore deposits.

Bulk sampling is recommended for concentrating grade ore deposits that require fine grinding to attain mineral liberation before concentration. Accurate data are needed concerning the various types and/or grades of ore present, and all aspects of the mining and processing of the ore materials to the final agglomerated product.

Bulk samples can be obtained in a number of ways, such as:

- (i) from the surface area by test pits,
- (ii) trenches, or small open cut mine pits, and
- (iii) from underground openings such as adits, shafts, tunnels, and drifts.

The validity of the bulk sample in regard to the purpose for which it is taken and the expected ore type is of paramount importance. The sample must be representative of the desired material. Because of the importance of the large-scale testing to several aspects of mine and plant design, there is small latitude for compromise or error.



■ SELECTION OF SAMPLING PROCEDURE

The usual mineral sampling methods have been discussed in earlier sections. Each method has its advantage and disadvantage. Therefore, some methods are very well-suited to some type of deposits. The process of matching a deposit with the best sampling procedure suited for it, requires certain criteria which are discussed below. Essentially, the criteria centre around the shape and type of mineralization of the deposit.

- (i) When the ore body is thick and the values of mineralization are uniformly distributed, sampling can be done by chip or grab sampling.
- (ii) When the ore body is of medium size and mineralization is uniform, a combined chip and channel sampling will give the best results.
- (iii) Where the ore body is too thin but occurs in benches or layers, sampling of various layers can be done by chip sampling.
- (iv) When a deposit is of very large dimensions, it becomes necessary to collect a large number of samples. In such cases, a large number of chip samples would give reliable results. Here, the quicker and not necessarily the most accurate method should be preferred.
- (v) With minerals like gold, rare metals, etc. where values are too spotty and irregular, bulk sampling would give the best results.
- (vi) Whenever the ore is banded, channel sampling would give the best results.

■ SIZE AND SPACING OF SAMPLES

One of the important steps in the design of a sample survey is to define the population to be sampled. For statistical application, mineral samples are usually in terms of weight or volume of mineral particles. Perhaps the most significant choice is to be made is about the sample size that will adequately represent the population under study and will allow us to determine precise enough estimates. Spatial interval between the samples influences the end results (namely grade and tonnage) considerably. A judicious decision is, therefore, to be taken about the number of samples and their positions in relation to one another so that an acceptable estimate about the population (the mineral deposit) can be arrived at without incurring over-expenditure and by a reasonable effort within a reasonable time.

Sample size is the number of observations used for calculating estimates of a given population. For example, if we analysed 30 random core-specimens of a given mineral deposit to assess its grade, '30 core-specimens' would be our sample size.

For most statistical applications, the total sample is in terms of number of items, attributes, or things, whereas mineral samples are usually in terms of weight or volume of mineral particles. While estimating mineral grade, the population may be a single stope within a mine, the particular zone within a complex ore body, or the entire mineral deposit.

Perhaps the most significant choice to be made is about the sample size that will adequately represent the population under study, and allow us to determine precise enough estimates.

Sample Spacing

If the results of feasibility study indicate financially viable project, the next step is to cost out the kind of detailed drilling program necessary to generate more reliable data. The most important consideration in this regard is drill hole spacing. It is a critical matter due to relationship between cost involved and the drill hole spacing. However, there are no rigid rules regarding spacing of sample channels in any type of deposit. The spacing can be controlled by mathematical analysis. Notably, it has been found that the formula using the standard error of statistical mean and the confidence interval is eminently suited for predetermining the channel spacing and the number of samples required for specific precision limit. For practical purposes, the sample is regarded as representing a certain zone surrounding the position of the sample, hence, sample should be taken in such a way that is it as representative as possible around it. For uniform deposits, each sample may be representative within a wide radius, but if the composition varies widely within very short intervals, then the sample represents only small radius. It will be apparent then that the space interval between samples will be governed by the degree of regularity of distribution of constituents throughout the mineral deposit.

Sampling Calculations Using Gy's Method

This method is a general-purpose calculation to determine the minimum size of sample needed to ensure that it will be representative of the whole lot, within specified limits.

Before using, approximate estimates of the following will be needed:

- The content of the species of interest in the lot (assay)
- The general shape of the particles
- The densities of the various species and phases present
- The particle size distribution
- The degree of liberation, and the grain size

Basic Equation:

$$\frac{WM}{W - M} = \frac{Cd_{\max}^3}{s^2}$$

Where:

- M= Minimum sample weight needed, grams
- W= Weight of the entire lot being sampled, grams
- C= Sampling constant for the material being sampled, g/cm³
- d_{max}=Dimensions of the largest pieces in the sample, cm
- s= value of the standard deviation that will be needed to give the desired level of assurance (assay units, such as % wt, ounces per ton, etc.)

When W is much larger than M, this equation can be simplified to:

$$M = \frac{Cd_{\max}^3}{s^2}$$

$$C = fglm$$

where: f= Shape factor
 g= Size distribution factor
 l= Liberation factor
 m= Composition factor

The shape factor, *f*, is taken to be equal to 0.5 for most common natural materials, but in cases where the material of interest is highly flattened and elongated, the shape factor is set at 0.2.

The size distribution factor, *g*, is determined by estimating the 95% passing size (*d*) and the 5% passing size (*d'*), and using the following guidelines:

- If $d/d' > 4$, then $g=0.25$;
- If $2 < d/d' < 4$, then $g=0.5$;
- If $d/d' < 2$, then $g=0.75$;
- If $d/d' = 1$ (monosized particles), then $g=1$.

The liberation factor, *l*, is a measure of the degree of dispersion of the valuable material through the bulk, and of the homogeneity of the material. It is calculated from the expression:

$$l = \sqrt{\frac{L}{d}}$$

where: L= the size where the values are essentially completely liberated (grain size), cm.

The composition factor, *m*, is calculated from the formula:

$$m = \left(\frac{1-a}{a} \right) ((1-a)r + at)$$

where: r= specific gravity of the valuable component
 t= specific gravity of the remainder of the material
 a= fractional average assay of the valuable substance (for example, if an iron-bearing slag is being analyzed, and the iron is present as Fe₂O₃, use the fraction of the material that is Fe₂O₃, not the fraction that is iron, and it would be expressed as a decimal fraction, not as a percent)

Sampling Calculations using Gaudin's Method

This method is limited to materials where the species of interest is a small fraction (a few percent or less) of the total volume. It does not allow for selecting the degree of certainty desired in the results, and it is not as generally useful or as precise as Gy's method. It would probably be most useful for calculating the size of a preliminary sample where minimal information about the material characteristics is available.

Basic Equations:

$$n = 0.45 \frac{x}{y^2}$$

$$n' = \frac{6}{\rho_s a^3}$$

$$S = \frac{n}{n'}$$

where:

x = volume fraction occupied by the species of interest

y = variation in volume fraction that would correspond to the maximum allowable error

n = number of particles that will be needed in the final sample

n' = estimated number of particles per gram

ρ_s = density of the bulk sample, in grams/cm³

a = top size of the material being sampled, in cm

S = total sample weight needed, in grams.

Bulk Material Parameters:

Material of Interest: CuFeS₂ in a silica matrix, 1.5% Cu (4.3318% CuFeS₂); Top Size= 1.5 cm;

CuFeS₂ grain size= 0.01 cm

Desired Sampling Accuracy: +/- 0.02% Cu, certainty of 0.99 (2.576 standard deviations)

CuFeS₂ specific gravity= 4.2; Overall specific gravity: 2.8; Broad size distribution

Calculations:

(Calculations were done using a spreadsheet, and were carried out to more digits than are shown here. Carrying out the calculations with the values as shown will give slightly different results due to roundoff variations.)

$f = 0.5$ (regular shape)

$g = 0.25$ ($d/d' > 4$ for a broad size distribution)

$$l = \sqrt{\frac{L}{d}} = \sqrt{\frac{0.01}{1.5}} = 0.081650$$

$a = 0.043318$, $r = 4.2$, and $t = 2.8$, and so:

$$m = \left(\frac{1 - 0.043318}{0.43318} \right) ((1 - 0.43318)4.2 + (0.043318)2.8) = 91.4177$$

$$C = fglm = (0.5)(0.25)(0.081650)(91.4177) = 0.933028$$

$$2.576s = \frac{0.02}{1.5} = 0.01333, \text{ so } s = 0.0051758$$

$$M = \frac{Cd^3}{s^2} = \frac{0.933028(1.5^3)}{0.0051758^2} = 117,539 \text{ grams of sample needed for 99\% certainty that the assay}$$

will be within +/- 0.02% Cu of the true assay.

Since the original data was only carried to 2 significant digits, this should be rounded to 120,000 grams of sample, or 120 kg.

Bulk Material Parameters:

Same material as for the previous example:

CuFeS₂, 1.5% Cu (4.3318% CuFeS₂); Top Size= 1.5 cm;

Desired Sampling Accuracy: +/- 0.02% Cu (+/- 0.0578% CuFeS₂)

CuFeS₂ specific gravity= 4.2; Overall specific gravity: 2.8

$a = 1.5 \text{ cm}$

$\rho_s = 2.8 \text{ gm/cm}^3$

$$x = \frac{\left(\frac{4.3318}{4.2} \right)}{\left(\frac{100}{2.8} \right)} = 0.028879$$

$$y = \frac{\left(\frac{0.0578}{4.2} \right)}{\left(\frac{100}{2.8} \right)} = 0.00038505$$

$$n = 0.45 \left(\frac{0.028879}{0.00038533^2} \right) = 87650$$

$$n' = \frac{6}{(2.8)(1.5^3)} = 0.63492$$

$$S = (87650)/(0.63492) = 138,050 \text{ grams}$$

(compared to 117,599 grams using Gy's method)

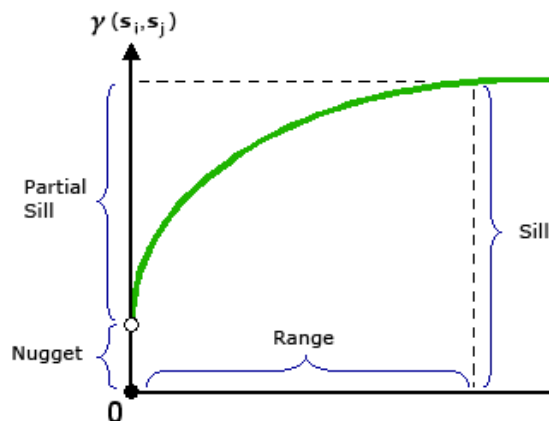
Again, this should be rounded to 2 significant digits, and so 140,000 grams, or 140 kg, of sample is required.

Application of statistical methods in sampling

The statistical concept of a sample is based on the fact that a random sample of size N will tend to have a distribution similar to the distribution of the universe being sampled. As N is increased, the distribution of the sample is increasingly likely to be similar to the distribution of the universe; and, when N includes all possible sample units, the distribution of the sample must be the same as the distribution of the universe.

Sample may be getting biased due to:

1. Getting sample from a zone of no interest
2. Improper weightage factor
3. Dilution of sample during collection
4. Taking sample of different sizes

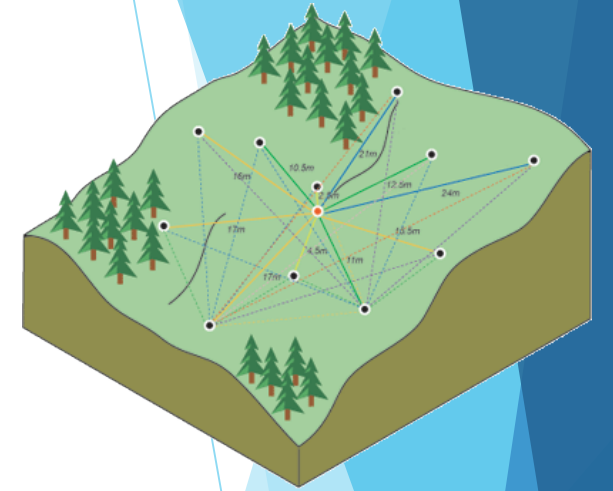


Variograms

A variogram is a plot showing how grade in a deposit varies versus distance. The function of ore grade variance is defined as γ (gamma) and is computed according to the following working relationship:

Kriging model

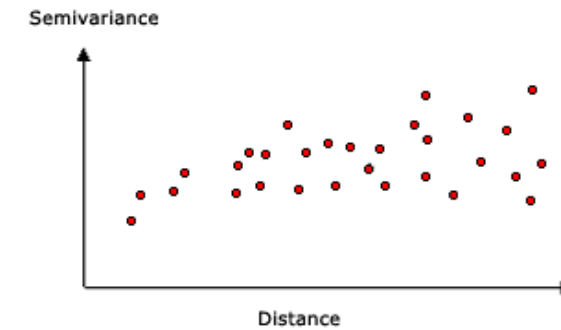
Kriging is a procedure to estimate the grade of a point or a block by using a weighted average of surrounding points



$$\hat{\gamma}_i(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [z(s_i) - z(s_i + h)]^2,$$

$$\lim_{h \rightarrow 0} \gamma(h) = C_0$$

where $N(h)$ is the number of pairs of points $z(s_i)$ and $z(s_i + h)$, separated by a vector h of which the two points s_i and s_{i+h} fall within the same stratum. Calculation of the variogram thus becomes insensitive to obvious differences in mean $z(s_i)$ values between different strata. The variogram is characterised by three major properties: the *nugget effect* (C_0), the *sill* (C) and the *range* (a). The nugget effect C_0 is a discontinuity of the variogram close to the origin:



and expresses both the variability at a scale smaller than the sampling interval and non-spatial variation. The nugget effect cannot be removed by close sampling, but only by repeated measurements (Trangmar *et al.*, 1985). The range is the lag distance where the variance approaches an asymptotic maximum, the sill. It expresses the distance beyond which samples are uncorrelated.

Once the variogram is known, the value of an attribute at any point in a mapping unit can be predicted from the available data points using kriging. Kriging gives the standard deviation of the prediction error as well. This standard deviation depends only on the variogram, the number of data and the spatial configuration with which these are taken (Burrough, 1991) and can hence be used to determine the required number of data.

KRIGING Model

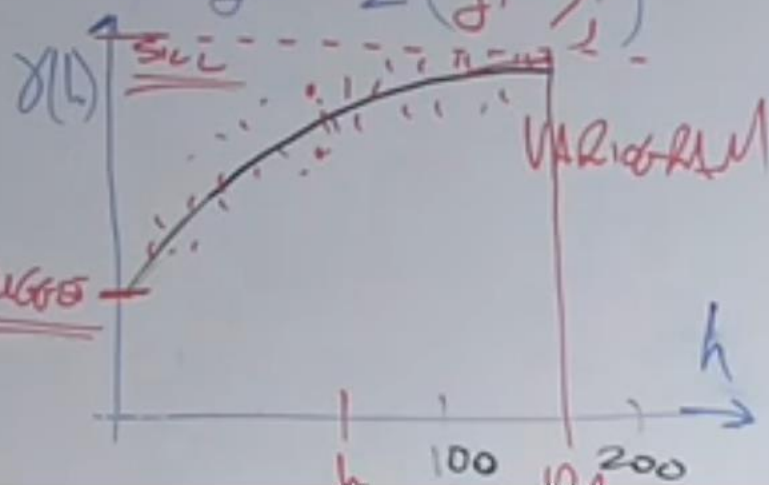
ASSUMPTIONS

THE Model

$$(x_i, y_i) \quad y_{\text{new}} = \omega^T y + \varepsilon_{\text{new}}$$

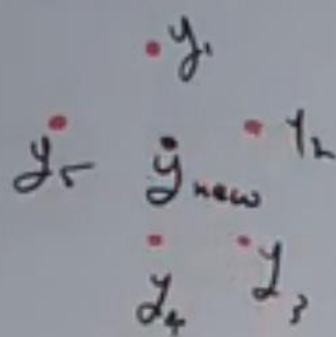
$$= \omega_1 y_1 + \omega_2 y_2 + \dots + \omega_n y_n + \varepsilon_{\text{new}}$$

$$\gamma(x_i, x_j) = \frac{1}{2} (y_i - y_j)^2$$



$$A\omega = b$$

$\gamma(x_i, x_j)$ (pointing to A)
 $\gamma(x_{\text{new}}, x_i)$ (pointing to b)



Thanks